

torch.cumsum#

<https://pytorch.org/docs/stable/generated/torch.cumsum.html#torch.cumsum>

Description:

Returns the cumulative sum of elements of input in the dimension dim. For example, if input is a vector of size N, the result will also be a vector of size N, with elements. input (Tensor) – the input tensor. dim (int) – the dimension to do the operation over dtype (torch.dtype, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None.

Function Signature

```
torch.cumsum(input, dim, *, dtype=None, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

`dtype` (`torch.dtype`, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to `dtype` before the operation is performed. This is useful for preventing data type overflows. Default: `None`. `out` (`Tensor`, optional) – the output tensor.

Code Examples

```
>>> a = torch.randint(1, 20, (10,))
>>> a
tensor([13,  7,  3, 10, 13,  3, 15, 10,  9, 10])
>>> torch.cumsum(a, dim=0)
tensor([13, 20, 23, 33, 46, 49, 64, 74, 83, 93])
```

Detailed Documentation

Documentation

Returns the cumulative sum of elements of input in the dimension dim.

For example, if input is a vector of size N, the result will also be a vector of size N, with elements.

input (Tensor) – the input tensor.

dim (int) – the dimension to do the operation over

dtype (torch.dtype, optional) – the desired data type of returned tensor. If specified, the input tensor is casted to dtype before the operation is performed. This is useful for preventing data type overflows. Default: None.

out (Tensor, optional) – the output tensor.